PH 716 Applied Survival Analysis

Part 4: Accelerated Failure Time Model

Zhiyang Zhou (zhou67@uwm.edu, zhiyanggeezhou.github.io)

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Motivating example 1: accelerated life testing of motorettes

Figure 1: Motorette in the Gedee Car Museum, Coimbatore, TN, India (From Wikipedia)

We often cannot wait years to observe failures under normal use. Instead, we run accelerated life tests by increasing stress (e.g., temperature) to make failures happen sooner, then model how stress changes the time-to-failure distribution.

The `MASS::motors` dataset contains an accelerated life testing of 10 motorettes at each of four temperatures. The recorded `time` is the time (hours) to failure, or censoring at 8064 hours.

```r
head(MASS::motors)
```

```
##   temp time cens
## 1  150 8064    0
## 2  150 8064    0
## 3  150 8064    0
## 4  150 8064    0
## 5  150 8064    0
## 6  150 8064    0
```

Scientific questions:

- Does higher temperature accelerate failure time (shorten lifetime)?
- If so, by how much?

Motivating example 2: capacitor failure under different stress levels

Capacitors are tested under various stress conditions (such as voltage or temperature) to study time to failure.

The dataset `survival::capacitor` contains failure times of capacitors tested under different temperature and voltage levels. Some units did not fail during the testing period, resulting in right-censored observations.

```r
head(survival::capacitor)
```

Figure 2: Capacitors (From Wikipedia)

```
##   temperature voltage fail time status
## 1         170     200   NA  439      1
## 2         170     200   NA  904      1
## 3         170     200   NA 1092      1
## 4         170     200   NA 1105      1
## 5         170     250   NA  572      1
## 6         170     250   NA  690      1
```

Scientific questions:

- Does higher voltage accelerate failure time?
- By what factor does lifetime change under increased stress?

Assumptions

- log-linear model: $\ln T_i = \beta_0 + \sum_{j=1}^p x_{ij}\beta_j + \sigma\varepsilon_i$
  - (Error scale) $\sigma > 0$
  - $\beta_j \in \mathbb{R}$
  - ε_i are iid with a specified distribution (e.g., normal, Gumbel, etc.)
- Equiv. $T_i = \exp(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j + \sigma\varepsilon_i)$
  - (Why is called "accelerated failure time model"?) The effect of covariates acts multiplicatively on the survival time and accelerates or decelerates the progress along the time axis.

Survival function

- If $\varepsilon_i \overset{\text{iid}}{\sim} N(0,1)$,
  - $S_{T_i}(t) = \Pr(\ln T_i > \ln t) = \Pr\{\varepsilon_i > \sigma^{-1}(\ln t - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j)\} = 1 - \Phi\{\sigma^{-1}(\ln t - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j)\}$
    - $\Phi(\cdot)$: the cdf of $N(0,1)$
  - i.e., $T_i \sim$ log-normal$(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j, \sigma^2)$
- If $\varepsilon_i \overset{\text{iid}}{\sim}$ the standard Gumbel distribution for minimum (i.e., $F_{\varepsilon_i}(\epsilon) = 1 - \exp(-\exp\epsilon)$),
  - P.S. $\min(X_1, X_2, \ldots, X_n) - \ln n \approx$ standard Gumbel distribution (for minimum) as $n \to \infty$ if $X_i \overset{iid}{\sim} \exp(1)$
  - $S_{T_i}(t) = \Pr\{\varepsilon_i > \sigma^{-1}(\ln t - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j)\} = 1 - F_{\varepsilon_i}\{\sigma^{-1}(\ln t - \beta_0 - \sum_{j=1}^p x_{ij}\beta_j)\} = \exp[-t^{1/\sigma}\exp\{-(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j)/\sigma\}] = \exp[-\{t/\exp(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j)\}^{1/\sigma}]$
  - i.e., $T_i \sim$ Weibull with $1/\sigma$ as the "shape" and $\exp(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j)$ as the "scale"
    - Widely used in practice, with a hazard descending ($1/\sigma < 1$), ascending ($1/\sigma > 1$), or flat ($1/\sigma = 1$) with respect to t
    - Specifically, $T \sim$ exponential with a hazard constant if $\sigma = 1$

Model fitting (Likelihood-based; for uncensored data)

- Observed event times $t_1, \ldots, t_n$ (no censoring for now)

- Construction of the likelihood function
  - Joint pdf of $T_1, \ldots, T_n$: $f_{T_1,\ldots,T_n}(t_1, \ldots, t_n; \boldsymbol{\theta})$
    - $\boldsymbol{\theta}$: a vector of unknown parameters
  - Likelihood function $L(\boldsymbol{\theta}) = f_{T_1,\ldots,T_n}(t_1, \ldots, t_n; \boldsymbol{\theta})$
    - $L(\boldsymbol{\theta}) = \prod_{i=1}^n f_{T_i}(t_i; \boldsymbol{\theta})$ if T_i are independent across i
- Estimation (maximum likelihood estimator, MLE)
- Inference
  - Hypothesis testing
  - Confidence interval

Model fitting (Likelihood-based; for right-censored data)

- Observed $\widetilde{T}_i = \tilde{t}_i$ and $\Delta_i = \delta_i$ (event indicator),
  - $\widetilde{T}_i$: the smaller one between T_i (event time) and C_i (right-cencoring time)
  - Assuming the independence across i
  - Assuming the independent and noninformative censoring, i.e.,
    - $T_i \perp C_i$ (conditional on covariates)
    - $S_{T_i}(t \mid \boldsymbol{\theta})$ and $S_{C_i}(t \mid \boldsymbol{\eta})$ have NO common parameter
- Joint pdf of $\widetilde{T}_i$ and Δ_i: $f_{\widetilde{T}_i,\Delta_i}(\tilde{t}_i, \delta_i) =$
  - $f_{T_i}(\tilde{t}_i \mid \boldsymbol{\theta}) S_{C_i}(\tilde{t}_i \mid \boldsymbol{\eta})$ if $\delta_i = 1$
  - $S_{T_i}(\tilde{t}_i \mid \boldsymbol{\theta}) f_{C_i}(\tilde{t}_i \mid \boldsymbol{\eta})$ if $\delta_i = 0$
    - Because
      - $\Pr(\widetilde{T}_i > t, \Delta_i = 1) = \Pr(C_i \geq T_i, T_i > t) = \int_t^\infty \Pr(C_i \geq u, T_i = u)\mathrm{d}u = \int_t^\infty S_{C_i}(u \mid \boldsymbol{\eta}) f_{T_i}(u \mid \boldsymbol{\theta})\mathrm{d}u$
      - $\Pr(\widetilde{T}_i > t, \Delta_i = 0) = \Pr(T_i \geq C_i, C_i > t) = \int_t^\infty \Pr(T_i \geq u, C_i = u)\mathrm{d}u = \int_t^\infty S_{T_i}(u \mid \boldsymbol{\theta}) f_{C_i}(u \mid \eta)\mathrm{d}u$
- Likelihood function: $L(\boldsymbol{\theta}, \boldsymbol{\eta}) = \prod_{i=1}^n f_{\widetilde{T}_i,\Delta_i}(\tilde{t}_i, \delta_i) = \prod_{i=1}^n \{f_{T_i}(\tilde{t}_i \mid \boldsymbol{\theta}) S_{C_i}(\tilde{t}_i \mid \boldsymbol{\eta})\}^{\delta_i} \{S_{T_i}(\tilde{t}_i \mid \boldsymbol{\theta}) f_{C_i}(\tilde{t}_i \mid \boldsymbol{\eta})\}^{1-\delta_i}$
  - Reducing to $\prod_{i=1}^n f_{T_i}(\tilde{t}_i \mid \boldsymbol{\theta})^{\delta_i} S_{T_i}(\tilde{t}_i \mid \boldsymbol{\theta})^{1-\delta_i} = \prod_{i=1}^n \lambda_{T_i}(\tilde{t}_i \mid \boldsymbol{\theta})^{\delta_i} S_{T_i}(\tilde{t}_i \mid \boldsymbol{\theta})$ if we are only concerned about the MLE of $\boldsymbol{\theta}$
- MLE of $\boldsymbol{\theta}$
- Inference
  - Hypothesis testing
  - Confidence interval

Model fitting (Likelihood-based; for general censored data)

- Assuming the independence across i and independence and noninformative censoring
- Likelihood function:

$$\prod_{i \in \mathfrak{D}} f_{T_i}(\tilde{t}_i) \prod_{i \in \mathfrak{R}} S_{T_i}(\tilde{t}_i) \prod_{i \in \mathfrak{L}} \{1 - S_{T_i}(\tilde{t}_i)\} \prod_{i \in \mathfrak{I}} \{S_{T_i}(\tilde{t}_{iL}) - S_{T_i}(\tilde{t}_{iR})\}$$

  - $\mathfrak{D}$: the set of **uncensored** subjects
  - $\mathfrak{R}$: the set of **right-censored** subjects
  - $\mathfrak{L}$ the set of **left-censored** subjects
  - $\mathfrak{I}$: the set of **interval-censored** subjects

- MLE
- Inference
  - Hypothesis testing
  - Confidence interval

Exponential regression for right-censored data

- Observed $\{\widetilde{T}_i = \tilde{t}_i, \Delta_i = \delta_i, x_{i1}, \ldots, x_{ip}\}$
  - $\widetilde{T}_i = \min(T_i, C_i)$
  - $\Delta_i = 1$ if $\widetilde{T}_i = T_i$ and zero if $\widetilde{T}_i = C_i$
- Assuming
  - Independence across i
  - Independent and non-informative censoring
  - $\ln T_i = \beta_0 + \sum_{j=1}^p x_{ij}\beta_j + \sigma\varepsilon_i$ with
    - $\varepsilon_i \overset{\text{iid}}{\sim} F_{\varepsilon_i}(\epsilon) = 1 - \exp(-\exp\epsilon)$
    - $\sigma = 1$
- Accordingly
  - T_i follows the exponential distribution with rate $\exp(-\beta_0 - \sum_{j=1}^p x_{ij}\beta_j)$
    - $\mathrm{E}(T_i) = \exp(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j)$
    - $\mathrm{median}(T_i) = (\ln 2)\mathrm{E}(T_i)$
    - $\lambda_{T_i}(t) = \exp(-\beta_0 - \sum_{j=1}^p x_{ij}\beta_j)$ for all $t > 0$
- Interpretation of parameters
  - β_0
    - $\exp(\beta_0)$: baseline expected survival time
    - $\exp(-\beta_0)$: baseline hazard rate
  - β_j, $j = 1, \ldots, p$ (holding all covariates other than the jth one)
    - In terms of survival time: A one-unit increase in the jth covariate multiplies the expected survival time by $\exp(\beta_j)$.
    - In terms of hazard rate
      - The hazard ratio associated with a one-unit increase in the jth covariate is $\exp(-\beta_j)$;
      - A one-unit increase in the jth covariate multiplies the hazard rate by $\exp(-\beta_j)$.
- Graphically check the correctness of model assumption
  1. Collect residuals $\ln T_i - \hat{\beta}_0 - \sum_j x_{ij}\hat{\beta}_j$ for uncensored subjects
  2. Compare residuals to a Gumbel random sample via the Q-Q plot.
- Difference in the outputs of `R` functions
  - Due to different ways of parameterization
  - `survival::survreg`: "Intercept" (i.e., $\hat{\beta}_0$) and $\hat{\beta}_j$, $j = 1, \ldots, p$
  - `flexsurv::flexsurvreg`: "rate" (i.e., $\exp(-\hat{\beta}_0)$) and $-\hat{\beta}_j$, $j = 1, \ldots, p$

Ex 4.2. Revisit the accelerated life testing of 10 motorettes

```r
head(MASS::motors)
data.ex42 = MASS::motors
data.ex42 = data.ex42[data.ex42$time != 0,] # ttr=0 not allowed in AFT models
is.factor(data.ex42$temp)
data.ex42$temp = factor(data.ex42$temp)
aft.ex42.1 = survival::survreg(
  survival::Surv(time, cens) ~ temp,
  data = data.ex42,
  dist="weibull",
  scale = 1, # error scale, i.e., sigma; not the "scale" for weibull distribution
  x = T
```

```r
)
summary(aft.ex42.1)
# Or
aft.ex42.2 = survival::survreg(
  survival::Surv(time, cens) ~ temp,
  data = data.ex42,
  dist="exponential"
)
summary(aft.ex42.2)
# Or using flexsurv::flexsurvreg
aft.ex42.3 = flexsurv::flexsurvreg(
  survival::Surv(time, cens) ~ temp,
  data = data.ex42,
  dist = "exponential"
)
aft.ex42.3

# The survival time for temp=150
rate = unname(exp(-aft.ex42.1$coefficients[1])) # rate of exp dist; temp=150 is the reference level
(ET= 1/rate) # expectation of T
(medT = log(2)*ET) # median of T
surv.fun = function(t){ # survival function
  return(
    1-pexp(t, rate = rate)
  )
}
curve(surv.fun, from = 0, to = 2*ET)

# The survival time for temp=220
rate = unname(exp( # rate of exp dist
  - aft.ex42.1$coefficients[1] - aft.ex42.1$coefficients['temp220']
))
(ET= 1/rate) # expectation of T
(medT = log(2)*ET) # median of T
surv.fun = function(t){ # survival function
  return(
    1-pexp(t, rate = rate)
  )
}
curve(surv.fun, from = 0, to = 2*ET)

# Graphically check the correctness of exponential assumption
set.seed(718)
g.rnd = ordinal::rgumbel(10000, max = F) # Gumbel random sample
lnTs.uncen = log(as.vector(data.ex42$time[data.ex42$cens==1]))
res = (
  lnTs.uncen - aft.ex42.1$x[data.ex42$cens==1,] %*% as.matrix(aft.ex42.1$coefficients)
)
qqplot(
  x = g.rnd,
  y = res,
  xlab = "Theoretical Quantiles",
  ylab = "Sample Quantiles"
```

```
)
qqline(res, distribution = function(p){ordinal::qgumbel(p,max=F)})
```

- Interpretation of $\hat{\beta}_1$
  - Compared with `temp=150`, `temp=170` multiplies the expected survival time by $\exp(\hat{\beta}_1)$;
  - The hazard at `temp=170` is $\exp(-\hat{\beta}_1)$ times the hazard at `temp=150`.

Weibull regression for right-censored data

- Observed $\{\widetilde{T}_i = \tilde{t}_i, \Delta_i = \delta_i, x_{i1}, \ldots, x_{ip}\}$
- Assuming
  - Independence across i
  - Independent and non-informative censoring
  - $\ln T_i = \beta_0 + \sum_{j=1}^p x_{ij}\beta_j + \sigma\varepsilon_i$ with
    - $\varepsilon_i \overset{\text{iid}}{\sim} F_{\varepsilon_i}(\epsilon) = 1 - \exp(-\exp\epsilon)$
- Accordingly
  - T_i follows the Weibull distribution with shape $1/\sigma$ and scale $\exp(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j)$
    - $E(T_i) = \exp(\beta_0 + x_{ij}\beta_j)\Gamma(1+\sigma)$
    - $\text{median}(T_i) = (\ln 2)^\sigma \exp(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j)$
    - $\lambda_{T_i}(t) = \sigma^{-1}t^{1/\sigma-1}\exp\{(-\beta_0 - \sum_{j=1}^p x_{ij}\beta_j)/\sigma\}$
- Interpretation of parameters
  - β_0: $\exp(\beta_0)$ is the baseline scale parameter of survival time.
  - β_j, $j \neq 0$ (after fixing all covariates other than the jth one): A one-unit increase in the jth covariate multiplies the survival time by $\exp(\beta_j)$.
    - Inconvenient to interpret β_j from the perspective of hazards (why?)
- Graphically check the correctness of model assumption
  1. Collect residuals $(\ln T_i - \hat{\beta}_0 - \sum_j x_{ij}\hat{\beta}_j)/\hat{\sigma}$ for uncensored subjects
  2. Compare residuals to a Gumbel random sample via the Q-Q plot.
- Difference in the outputs of `R` functions
  - `survival::survreg`: "Intercept" ($\hat{\beta}_0$), "scale" ($\hat{\sigma}$), "log(scale)" ($\ln\hat{\sigma}$), and $\hat{\beta}_j$, $j = 1, \ldots, p$
  - `flexsurv::flexsurvreg`: "shape" ($1/\hat{\sigma}$), "scale" ($\exp(\hat{\beta}_0)$), and $\hat{\beta}_j$, $j = 1, \ldots, p$

Ex 4.3. Revisit the accelerated life testing of 10 motorettes

```
head(MASS::motors)
data.ex43 = MASS::motors
data.ex43 = data.ex43[data.ex43$time != 0,] # ttr=0 not allowed in AFT models
is.factor(data.ex43$temp)
data.ex43$temp = factor(data.ex43$temp)
aft.ex43.1 = survival::survreg(
  survival::Surv(time, cens) ~ temp,
  data = data.ex43,
  dist="weibull",
  x = T
)
summary(aft.ex43.1)
# OR using flexsurv::flexsurvreg
```

```r
aft.ex43.2 = flexsurv::flexsurvreg(
  survival::Surv(time, cens) ~ temp,
  data = data.ex43,
  dist = "weibull"
)
aft.ex43.2

# The survival time for temp=150
shape = 1/aft.ex43.1$scale # shape of Weibull dist
scale = unname(exp( # scale of Weibull dist; temp=150 is the reference level
  aft.ex43.1$coefficients[1]
))
(ET = scale*gamma(1+1/shape)) # expectation of T
(medT = scale*(log(2))^(1/shape)) # median of T
surv.fun = function(t){ # survival function
  return(
    1-pweibull(t, shape = shape, scale = scale)
  )
}
curve(surv.fun, from = 0, to = 2*ET)

# The survival time for temp=220
shape = 1/aft.ex43.1$scale # shape of Weibull dist
scale = unname(exp( # scale of Weibull dist
  aft.ex43.1$coefficients[1]+
  aft.ex43.1$coefficients['temp220']
))
(ET = scale*gamma(1+1/shape)) # expectation of T
(medT = scale*(log(2))^(1/shape)) # median of T
surv.fun = function(t){ # survival function
  return(
    1-pweibull(t, shape = shape, scale = scale)
  )
}
curve(surv.fun, from = 0, to = 2*ET)

# Graphically check the correctness of weibull assumption
set.seed(718)
g.rnd = ordinal::rgumbel(10000, max = F) # gumbel random sample
lnTs.uncen = log(as.vector(data.ex43$time[data.ex43$cens==1]))
res = (
  lnTs.uncen - aft.ex43.1$x[data.ex43$cens==1,] %*% as.matrix(aft.ex43.1$coefficients)
)/aft.ex43.1$scale
qqplot(
  x = g.rnd,
  y = res,
  xlab = "Theoretical Quantiles",
  ylab = "Sample Quantiles"
)
qqline(res, distribution = function(p){ordinal::qgumbel(p,max=F)})
```

- Interpretation of $\hat{\beta}_1$: Compared with `temp=150`, `temp=170` multiplies the survival time by $\exp(\hat{\beta}_1)$.

Log-normal regression for right-censored data

- Observed $\{\widetilde{T}_i = \tilde{t}_i, \Delta_i = \delta_i, x_{i1}, \ldots, x_{ip}\}$
- Assuming
  - Independence across i
  - Independent and non-informative censoring
  - $\ln T_i = \beta_0 + \sum_{j=1}^p x_{ij}\beta_j + \sigma\varepsilon_i$ with
    - $\varepsilon_i \overset{\text{iid}}{\sim} N(0,1)$
- Accordingly
  - T_i follows the log-normal distribution with parameters $\beta_0 + \sum_{j=1}^p x_{ij}\beta_j$ and σ
    - $\mathrm{E}(T_i) = \exp(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j + \sigma^2/2)$
    - $\text{median}(T_i) = \exp(\beta_0 + \sum_{j=1}^p x_{ij}\beta_j)$
- Interpretation of parameters
  - β_0: $\exp(\beta_0)$ is the baseline median survival time.
  - β_j, $j \neq 0$ (after fixing all covariates other than the jth one): A one-unit increase in the jth covariate multiplies the mean/median survival time by $\exp(\beta_j)$.
- Graphically check the correctness of model assumption
  1. Collect residuals $(\ln T_i - \hat{\beta}_0 - \sum_j x_{ij}\hat{\beta}_j)/\hat{\sigma}$ for uncensored subjects
  2. Compare residuals to a $N(0,1)$ random sample via the Q-Q plot.
- Difference in the outputs of `R` functions
  - `survival::survreg`: "Intercept" ($\hat{\beta}_0$), "scale" ($\hat{\sigma}$), and $\hat{\beta}_j$, $j = 1, \ldots, p$
  - `flexsurv::flexsurvreg`: "meanlog" ($\hat{\beta}_0$), "sdlog" ($\hat{\sigma}$), and $\hat{\beta}_j$, $j = 1, \ldots, p$

Ex. 4.4. Capacitor failure under different stress levels

```r
data.ex44 = survival::capacitor
data.ex44 = data.ex44[data.ex44$time != 0,] # ttr=0 not allowed in AFT models
is.factor(data.ex44$temperature)
is.factor(data.ex44$voltage)
data.ex44$temperature = factor(data.ex44$temperature)
data.ex44$voltage = factor(data.ex44$voltage)
aft.ex44.1 = survival::survreg(
  survival::Surv(time, status) ~ temperature + voltage,
  data = data.ex44,
  dist="lognormal",
  x = T
)
summary(aft.ex44.1)
# OR using flexsurv::flexsurvreg
aft.ex44.2 = flexsurv::flexsurvreg(
  survival::Surv(time, status) ~ temperature + voltage,
  data = data.ex44,
  dist = "lognormal"
)
aft.ex44.2

# The survival time for temp=180 and voltage=300
sigma = aft.ex44.1$scale
```

```r
mu = aft.ex44.1$coefficients[1] +
  aft.ex44.1$coefficients['temperature180'] +
  aft.ex44.1$coefficients['voltage300']
(ET = exp(mu+sigma^2/2)) # expectation of T
(medT = exp(mu)) # median of T
surv.fun = function(t){ # survival function for treatment='pyridoxine'
  return(
    1-pnorm(log(t), mu = mu, sd = sigma)
  )
}
curve(surv.fun, from = 0, to = 2*ET) # plot the survival curve

# Graphically check the correctness of log-normal assumption
set.seed(718)
lnTs.uncen = log(as.vector(data.ex44$time[data.ex44$status==1]))
res = (
  lnTs.uncen - aft.ex44.1$x[data.ex44$status==1,] %*% as.matrix(aft.ex44.1$coefficients)
)/sigma
qqnorm(
  y = res,
  xlab = "Theoretical Quantiles",
  ylab = "Sample Quantiles"
)
qqline(res)
# Shapiro-Wilk test for normality
shapiro.test(res)
```

Ex. 4.5. Randomized trial of triple therapy vs. patch for smoking cessation: `asaur::pharmacoSmoking`

- The dataset `asaur::pharmacoSmoking` contains data from a randomized trial of triple therapy versus nicotine patch alone for smoking cessation in smokers with medical illnesses (Steinberg et al., 2009).
- Specific aim: To evaluate extended duration of a triple-medication combination versus therapy with the nicotine patch alone in smokers with medical illnesses.
- Fit the three AFT models to the data, and see which model fits the data better.
- Conclude on the effect of triple therapy vs. patch alone on the time to smoking cessation.

Pros and cons of AFT model

- Easy to interpret coefficients in terms of survival time
- Distribution assumptions may be too strong
- Can handle non-standard situations such as interval censoring
- Yields estimates of functions like hazard and survival for all times (even beyond the scope of follow-up)
  - But the extrapolation beyond the observed data range is not reliable